# Agent-Computer Observation Interfaces
# Enable Dynamic Computer Use

**Anonymous submission**


### Abstract

Computer-use (CU) agents commonly condition each decision on one screenshot and receive no audio input. This action-coupled observation scheme omits transient visual content and speech that occur between decisions. We treat the *observation interface*—when the environment is sampled, which modalities are captured, and how observations are represented and retained—as a system design dimension distinct from the underlying model and action space.

We introduce the **Agent-Computer Observation Interface (AOI)**, a perception layer that monitors the screen and system audio between decisions. AOI constructs a structured observation record from gated keyframes, volume-gated speech transcription, and model-generated visual narration that persists as text. It augments image-and-text CU models without retraining or changing their action spaces.

We evaluate AOI on **DynaCU-Bench**: 100 dynamic browser tasks across 10 families, plus a 50-task static control. Across eight CU models from 7B open-source models to frontier systems, AOI improves task success by 17–48 percentage points; Claude Sonnet 4.6 achieves a three-seed mean of 78.7% (3.1 pp standard deviation). On a 12-task audio subset, AOI-equipped Claude completes 12/12 tasks, compared with 3/12 for OpenAI Realtime and 0/12 for Gemini Live. Component analyses show that keyframes provide most of their value through visual narration and that the preferred observation channels depend on the underlying model.


## 1 Introduction

Most computer-use (CU) agents operate through a discrete observation–action loop: the model receives a screenshot, predicts an action, and receives another screenshot after execution (Xie et al. 2024; Zhou et al. 2023; He et al. 2024). Because observation occurs only at these action boundaries, transient content that disappears before the next screenshot never enters the trajectory. Such agents also miss voice reminders, meeting audio, podcasts, and spoken video content. Such failures can therefore reflect missing input rather than insufficient reasoning or an inadequate action space.

Prior work enriches individual screenshots (Zhou et al. 2023; Deng et al. 2023; Yang et al. 2023; Zheng et al. 2024), evaluates temporal reasoning in video environments (Chen et al. 2024; Jang et al. 2024; Zhang et al. 2025; Chen et al. 2025), or processes continuous streams (Google 2025; OpenAI 2024). These directions leave a distinct question: can an existing screenshot-based CU agent acquire and retain multimodal information between actions without retraining? We use *observation interface* for the sampling time, modalities, representation, and retention of information supplied to an agent. Following work on action interfaces (Yang et al. 2024), we treat observation as independently configurable from the model.

We introduce the **Agent-Computer Observation Interface (AOI)**, a layer between the environment and an image-and-text CU agent. Between decisions, a pixel-change gate retains informative keyframes and a volume gate triggers speech transcription. The CU model also narrates new visual information, which persists as text after older images are removed. AOI supplies these channels with the current screenshot in a structured record without changing the model parameters or action space.

We evaluate AOI using **DynaCU-Bench**, which contains 100 dynamic browser tasks across 10 families, together with a 50-task static control. The main evaluation covers six closed- and open-source CU models, with two Qwen3-VL models as an independent replication. AOI improves all eight by 17–48 percentage points without retraining. It completes 50/50 static-control tasks versus 43/50 for the screenshot baseline. On a 12-task audio subset, AOI-equipped Claude completes all tasks, versus 3 for OpenAI Realtime and 0 for Gemini Live.

Ablations show that both the structured record and new perceptual input contribute to the gains, that keyframes provide most of their value through narration, and that the best channel combination is model-dependent.

**Contributions.**

1. We formulate the observation interface through sampling time, modality, representation, and retention, and develop AOI to decouple observation from action.
2. We introduce DynaCU-Bench: 100 dynamic browser tasks across 10 families and a 50-task static control.
3. Across eight CU models and two streaming baselines, we identify narration and model-specific channel selection as key factors (Sections 5–7).

## 2 Background: The CU Agent Loop

Computer-use agent architectures and evaluation are surveyed by Sager et al. (Sager et al. 2025). Most screenshot-based CU agents follow a discrete *observe* $\rightarrow$ *reason* $\rightarrow$ *act* cycle: capture screenshot $s_t$, sample and execute action $a_t$ conditioned on $s_t$ and history $\tau$, wait a short buffer $\delta$, and repeat. The **action space** $\mathcal{A}$ is a small grammar of commands—click, type, scroll, wait, done—(Yang et al. 2024). By contrast, the **observation space** $\mathcal{S}$ remains severely limited to a single RGB screenshot per step.

This design leaves agents blind between screenshots. During the typical 3–5 second interval, agents miss most motion and transient UI events. Furthermore, the lack of an audio channel leaves agents deaf to all sounds. Concretely, four categories of content are systematically missed: (i) *transient events*—toasts and dialogs that appear and auto-dismiss within a single interval; (ii) *continuous visual media*—video, animated tutorials, and transitioning slides; (iii) *audio*—meeting speech, notification sounds, and spoken prompts; and (iv) *periodic noise*—spinners and blinking cursors that should be *ignored* yet disrupt naive frame differencing methods. AOI expands $\mathcal{S}$ to cover (i)–(iii) while suppressing (iv), all without modifying the model or $\mathcal{A}$.

## 3 System Design

AOI is a model-agnostic perception layer between the environment and an existing CU model. It observes between agent steps and supplies keyframes, audio transcripts, and persistent visual narrations in standard image–text format. It requires no retraining, but adds latency and therefore targets human-paced rather than real-time use.

### 3.1 Architecture Overview

AOI has three components (Figure 1). A sub-millisecond pixel-change gate skips visual processing on static content, and an RMS energy gate skips transcription during silence. For changed frames, CLIP-ViT-B/16 adds approximately 7 ms; audio processing is dominated by Whisper large-v3. The CU model generates visual narration when new visual observations are supplied.

(1) **Inter-step keyframe capture**: Samples at $\sim$3 Hz and selects up to five frames per step using pixel-change and CLIP-semantic gates.

(2) **Volume-gated audio observer**: RMS energy gate followed by Whisper large-v3 transcription, active only when speech is detected.

(3) **Visual narration context**: The CU model itself generates a brief description of new visual content as a side-output of each inference call. Narrations persist as text after images are pruned.

### 3.2 Inter-Step Keyframe Capture

We sample at 3 Hz and compare 64×64 grayscale frames. A pixel changes when intensity differs by more than 10/255. We skip samples below 1% changed pixels and otherwise capture when CLIP cosine distance from the anchor is at least 0.04 or changed pixels reach 3% (Algorithm 1); we retain at most five frames per step. We

---

Algorithm 1: Adaptive keyframe extraction used in the main experiments.

**Require:** Screen sample $x_t$, anchor embedding $e_{\text{anchor}}$, pixel thresholds $\alpha, \beta$, CLIP threshold $\theta$

$\Delta_{\text{px}} \leftarrow$ PixelChangeRatio$(x_t, x_{\text{prev}})$
**if** $\Delta_{\text{px}} < \alpha$ **then**
  **return** SKIP {Stage 1: no pixel change}
**end if**
$e_t \leftarrow$ CLIP$(x_t)$
$d \leftarrow 1 - \cos(e_t, e_{\text{anchor}})$
**if** $d \geq \theta$ **or** $\Delta_{\text{px}} \geq \beta$ **then**
  $e_{\text{anchor}} \leftarrow e_t$ {Re-anchor}
  **return** CAPTUREKEYFRAME$(x_t, t)$
**end if**
**return** SKIP {Below both capture thresholds}

---

chose 0.04 from $\{0.001, 0.01, 0.04, 0.05, 0.08, 0.10, 0.15\}$ on a transient-dialog calibration page: it captured appearance and disappearance (0.0805 and 0.0796) while rejecting static variation (0.0006). Changed-pixel ratios were 0 for static frames, $\sim$0.93 for both dialog transitions, and 0.0015 for a small UI shift, supporting the 0.01 skip gate. A later task-level sensitivity sweep evaluated CLIP thresholds $\{0.02, 0.04, 0.08, 0.12, 0.20, 0.30\}$. We did not preserve complete development ranges or selection criteria for the remaining parameters.

### 3.3 Volume-Gated Audio Observation

We capture 16 kHz mono PulseAudio and invoke Whisper large-v3 (Radford et al. 2023) at RMS energy $\geq$ 0.01. A 70 s buffer supplies 5 s recent and 60 s rolling transcripts, refreshed at most every 5 s. We configure Whisper with beam size 1, best-of 1, English, 300 ms minimum VAD silence, and 200 ms speech padding. We transcribe speech only. We did not record the ASR device; our service default was GPU float16, with CPU int8 optional.

### 3.4 Visual Narration for Long-Term Context

CU models retain only the most recent images in context, pruning earlier keyframes. Consequently, on tasks that span many steps, the agent loses all visual memory of prior content.

To address this, in the same inference call that produces an action, the CU model generates a brief narration of new visual information. Narrations remain stored after images are pruned, but the model receives only the five most recent step records at each decision. Following the caption-then-reason principle of LLoVi (Zhang et al. 2023), narrations are generated by the CU model itself and are task-relevant rather than generic.

### 3.5 Observation Record

Before decision $t$, the model receives a structured record combining persistent text context (prior transcripts, narrations, and actions) with observations collected after action $t-1$ (new audio transcription, keyframes if any, and the latest screenshot). The structured scaffold also includes a compact list of DOM elements available for interaction. On static

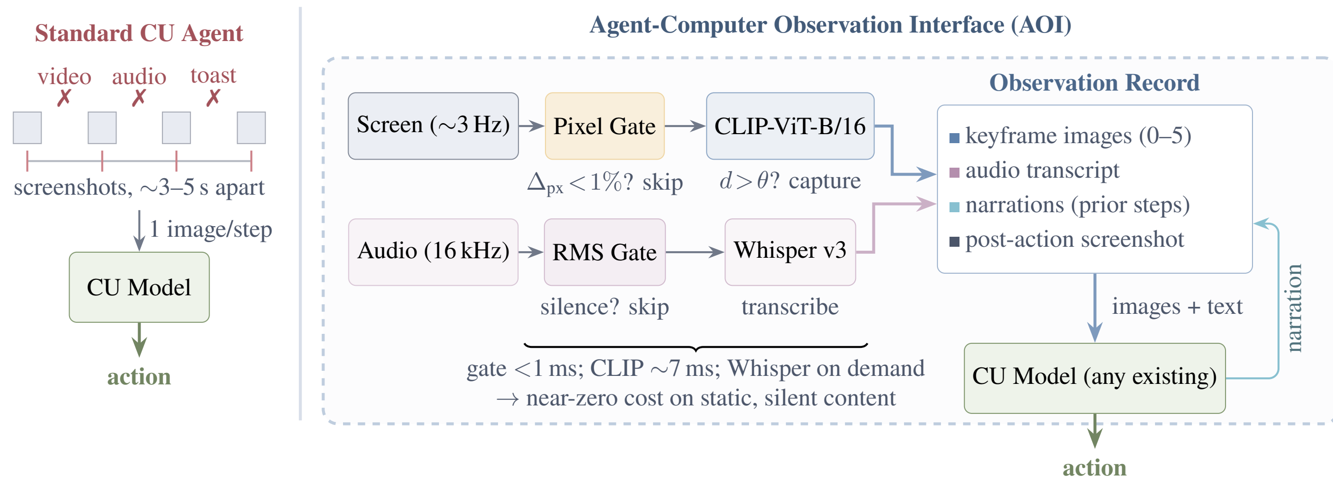


Figure 1: Standard CU agents vs. AOI-equipped agents. *Left:* Standard agents observe only periodic screenshots and therefore miss audio and inter-step visual events. *Right:* AOI continuously monitors screen and audio streams, selects informative observations, and supplies them to the underlying CU model in a structured record. Textual narrations and transcripts persist after older images are removed.

**Observation Record — Step $N$ (Meeting task)**

```
# CONTEXT -- text from prior steps
(no images)
Step N-1 (t ≈ 18.5-22.0 s):
AUDIO: "Here's the team. As you can
see we
have strong cross-functional
coverage."
VISUAL: "Meeting slide shows Project
Team
with 6 members listed in a grid."
ACTION: wait()
# NEW -- current interval
observations
Step N (t ≈ 22.0-25.5 s):
AUDIO: "And I want to mention, we've
moved
the launch date to April 28th.
Please update your calendars."
[IMAGE: post-action screenshot]
# TASK
Read the meeting slides and listen
to the
discussion. What is the new launch
date?
Enter it in the Launch Date field.
```

Figure 2: Example observation record at step $N$ of a meeting task. Prior observations persist as text, while the current step contributes a fresh audio transcription and screenshot. In this example, no keyframes are included because the slide does not change.

tasks, the perceptual input reduces to the latest screenshot. Figure 2 shows an example.

### 3.6 Implementation

We ran experiments on one Linux host with an NVIDIA RTX PRO 6000 Blackwell GPU (96 GB VRAM) and 192 GB RAM. We used Python 3.11; Playwright 1.49/headless Chromium; PulseAudio 16.1 with `parecord`/`pacat`; ffmpeg; edge-TTS; CLIP ViT-B/16; Whisper large-v3 via faster-whisper/CTranslate2; and vLLM 0.19.0. We ran CLIP and local models on the GPU, called cloud models through HTTP APIs, and used a 4,096-token local context. We did not preserve the CPU/core count, Linux release, CUDA/driver versions, remaining stack versions or revisions, or a lockfile.

## 4 DynaCU-Bench

### 4.1 Design Principles

Existing CU benchmarks (Xie et al. 2024; Zhou et al. 2023; He et al. 2024) focus almost exclusively on static tasks. We introduce **DynaCU-Bench**, a 100-task benchmark of realistic browser-based tasks that require dynamic visual or audio perception.

The benchmark follows five design principles: tasks must be unsolvable from static screenshots, reflect real computer use, run reproducibly in a headless browser, cover audio, visual-temporal, and interaction axes, and include difficulty stratification.

It spans 10 task families (Table 1). Each family contains 10 tasks with balanced difficulty (3 easy, 4 medium, 3 hard).

### 4.2 Evaluation Protocol

Each task runs in a headless Chromium browser with virtual audio devices for playback and microphone injection; spoken content is rendered using high-quality text-to-speech. Most tasks (93) are scored by a deterministic check on the resulting page state. The remaining tasks either combine that check

| Family | Axes | Description |
|---|---|---|
| **Podcast** | AUD | Listen to spoken narration; extract facts or answer questions |
| **Meeting** | AUD+VIS+INT | Follow a meeting with slides and speech; take notes or act |
| **Screencast** | VIS | Watch terminal/IDE recordings with auto-advancing frames |
| **Carousel** | VIS | Perceive CSS transitions, rotating content, product carousels |
| **Dashboard** | VIS | Monitor live-updating dashboards |
| **Transient** | VIS | Respond to toasts and auto-dismissing banners |
| **Phone** | AUD+INT | Listen to a synthesized voice call and respond |
| **Interview** | AUD+INT | Answer spoken questions; complete interviews |
| **Collab** | VIS+INT | Handle remote edits in collaborative documents |
| **Games** | VIS+INT | Play interactive games requiring temporal attention |

Table 1: DynaCU-Bench task families and axes: AUD denotes audio, VIS visual-temporal perception, and INT real-time interaction. Each family has 3 easy, 4 medium, and 3 hard tasks.

| Model | Standard | AOI full | Δ |
|---|---|---|---|
| 235B-A22B | 22 | **64** | +42 |
| 30B-A3B | 18 | **42** | +24 |

Table 2: Open-source replication on DynaCU-Bench (100 tasks) using Qwen3-VL models. Results use the same evaluation protocol as Figure 3.

with an LLM quality rubric (6) or rely solely on the LLM judge (1). A task succeeds when its final binary score is 1. For LLM-scored tasks, we use Gemini 2.0 Flash to assign a score in $[0, 1]$ from the task-specific rubric; scores of at least 0.5 pass, and hybrid tasks must also pass their DOM check. We use provider-default judge decoding and do not repeat or cross-validate judge calls. We use binary completion because it directly measures whether the agent fulfills the user-facing objective. Per-family success exposes modality-specific failures hidden by the aggregate; steps, tokens, cost, and wall-clock time are secondary efficiency measures rather than substitutes for completion. A run ends after 15 agent steps or its wall-clock budget. The budget is the stimulus duration plus 10 s; non-standard modes receive a fixed additional 30 s for observation processing. Errors, invalid outputs, and timeouts count as failures. These budgets bind only for the slowest models (e.g. Grok-4 at ∼80–180 s/call; Section 7).

# 5 Evaluation

## 5.1 Setup

**CU models.** We evaluate Claude Sonnet 4.6 (`claude-sonnet-4-6`) (Anthropic 2024a), GPT-5.4 (`gpt-5.4`) (OpenAI 2026), Gemini 2.5 Flash (`gemini-2.5-flash`) (Google DeepMind 2025), Grok-4 (`grok-4`) (xAI 2026), EvoCUA-32B (`meituan/EvoCUA-32B-20260105`) (Meituan 2026), and Fara-7B (`microsoft/Fara-7B`) (Awadallah et al. 2025). For the OpenRouter replications, we use `qwen/qwen3-vl-235b-a22b-instruct` and `qwen/qwen3-vl-30b-a3b-instruct` (Qwen Team 2025) (Table 2). We cap every non-streaming response at 1,024 tokens and omit temperature, top-$p$, and API seeds, using provider/runtime defaults. We use a 4,096-token context for local vLLM models and float16 for Fara. We did not record EvoCUA dtype/quantization consistently, and we did not retain hosted snapshot revisions or OpenRouter's routed provider.

**Observation configurations.** The primary contrast is between **Standard** (one screenshot per step, no audio—the current paradigm) and **AOI full** (inter-step keyframes + audio transcription + visual narration). We evaluate both modes across all six models. All remaining modes are ablations and diagnostic controls run on Claude Sonnet 4.6 (unless otherwise noted).

## 5.2 Main Results

For example, on meeting tasks the standard agent often fails because it cannot hear spoken answers, while AOI transcribes the audio and completes the task in far fewer steps. On video and carousel tasks, narration often enables correct answers in a single step.

*Statistical protocol.* Unless stated otherwise, each configuration is one run over 100 binary tasks. For a success count $k$ among $n$ tasks, we use a 95% Wilson interval with $z = 1.96$: center $(\hat{p} + z^2/2n)/(1 + z^2/n)$ and half-width $z\sqrt{\hat{p}(1-\hat{p})/n + z^2/(4n^2)}/(1+z^2/n)$. For paired configurations, let $b, c$ be their discordant success counts, $N = b+c$, and $K = \min(b, c)$. We report the two-sided exact McNemar value $p = \min\{1, 2\sum_{i=0}^{K}\binom{N}{i}2^{-N}\}$. For the six planned Standard–AOI comparisons in Figure 3, we control family-wise error at $\alpha = 0.05$ with Holm's step-down correction. All six remain significant (raw $p = 2.9\times10^{-12}$–$4.9\times10^{-4}$; maximum Holm-adjusted $p = 4.9\times10^{-4}$). Component and per-family tests are exploratory and uncorrected.

AOI improves every model by 17–48 points (Figure 3). Claude Sonnet 4.6 reaches 82%; Gemini 2.5 Flash gains 48 points; EvoCUA-32B rises from 18% to 55%; and Fara-7B from 17% to 34%. Magnitude and optimal channels remain model-dependent.

**Additional open-source replication (Qwen3-VL family).** Both Qwen3-VL models reproduce the same trend: the 235B model improves by +42 pp (22%→64%) and the 30B

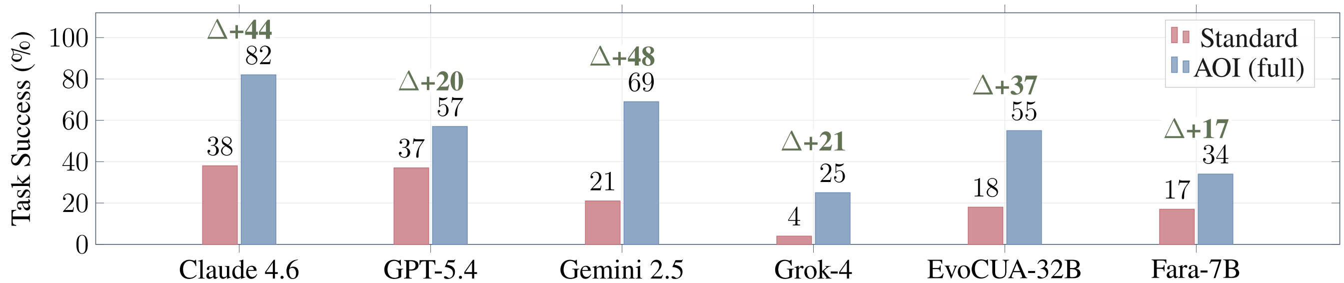


Figure 3: Main results on DynaCU-Bench (100 tasks). AOI substantially improves task success across all evaluated CU models without retraining. Green labels indicate the absolute improvement over the standard screenshot baseline.

model by +24 pp (18%→42%), reinforcing that AOI's gains are model-agnostic.

**Variance.** Repeating the headline configuration (Claude + AOI full) across three seeds yields 82%, 78%, 76% (mean 78.7%, std 3.1 pp). The reported 82% thus sits at the upper end of a ∼3 pp decoding variability band. This quantifies absolute score variance and is complementary to the paired McNemar tests reported above.

## 5.3 Per-Family Analysis

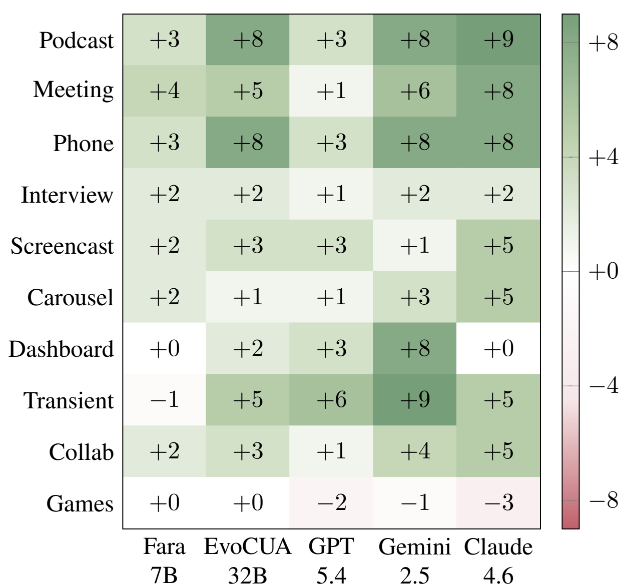


Figure 4: Per-family AOI gain (AOI full − Standard, tasks per 10) across five models. Families are grouped by modality. Green indicates improvement; red indicates regression.

Figure 4 reveals three patterns. Audio tasks (Podcast, Phone, Meeting) improve the most because they are impossible from screenshots alone. Visual-temporal tasks (Screencast, Carousel, Transient) also improve consistently, whereas Dashboard gains are modest because the relevant information is often visible in a single screenshot. Games is the only family that consistently regresses: it is limited by action latency rather than perception, and AOI's additional observation time exceeds its benefit.

## 5.4 Difficulty Breakdown

AOI improves performance across all difficulty tiers, with the largest absolute gains on medium and hard tasks for stronger models. On Fara-7B, improvements are concentrated on easy tasks, suggesting that reasoning rather than perception is the dominant remaining bottleneck.

## 5.5 Efficiency

Better perception reduces the number of interaction steps enough to outweigh the richer observations, lowering token costs by 15–50% across cloud models. Wall-clock time decreases only for the slowest models, where fewer agent steps dominate the added observation overhead. AOI therefore improves cost efficiency while trading off latency on faster models.

# 6 Decomposing the Gain

We decompose AOI's gains in two stages: separating the structured prompt-format effect from genuine perception improvements, then breaking down the perception component channel by channel.

## 6.1 Prompt Format vs. Perception

We first rule out the possibility that AOI helps only through prompt reformatting.

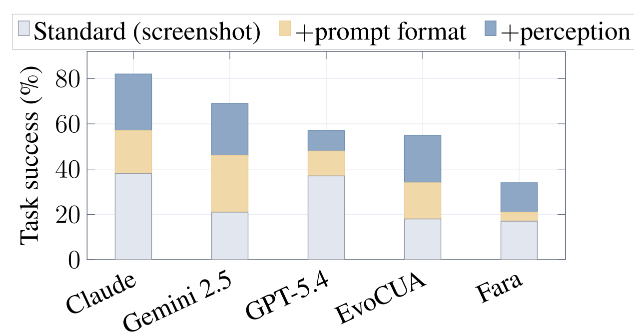


Figure 5: Bars stack the standard screenshot baseline, the gain from the structured observation record alone, and the additional gain from perception (keyframes, audio, and narration).

On the purely static **Static-50** benchmark, AOI achieves 50/50 while the screenshot baseline scores 43/50. Because no dynamic observations are available, the observed gain is attributable to the structured prompt format; AOI causes no failures on this task set.

On dynamic tasks, a control using only AOI's structured observation record (no keyframes or audio) isolates the prompt-format contribution. Figure 5 shows both prompt-format and perception components are positive for every model. The balance is model-specific: Claude is perception-heavy, Gemini 2.5 is prompt-heavy, and Fara-7B cannot exploit the richer prompt. Among elements of the structured scaffold, the DOM-element list introduced in Section 3.5 contributes the largest gain.

### 6.2 Perception Channel Contributions

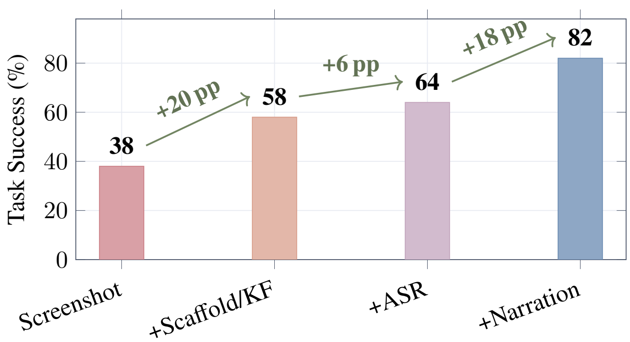


Figure 6: Progressive contribution of AOI components on Claude Sonnet 4.6.

| Comparison | Total |
|---|---|
| Standard → Uniform 1 FPS | 38 / 58 |
| Uniform 1 FPS → Uniform 3 FPS | 58 / 56 |
| Uniform 3 FPS → Pixel-diff | 56 / 58 |
| Pixel-diff → Random keyframes | 58 / 56 |
| Random keyframes → CLIP adaptive | 56 / 58 |
| CLIP adaptive → AOI visual+ASR | 58 / 64 |
| AOI visual+ASR → AOI full | 64 / 82 |

Table 3: Results for successive ablation configurations on Claude Sonnet 4.6. Total reports successful tasks out of 100.

Figure 6 shows incremental gains. Keyframe images add only +1 pp as raw input but +10 pp once narrated (Figure 7b), with benefits concentrated on Transient and Carousel tasks. Table 3 reports results across keyframe selection strategies. Audio transcription adds directional gains on audio families, while visual narration is the largest single component (+18 pp).

A narration-discarded control (Figure 7a) attributes most of narration's benefit to inference-time articulation, with persistence mattering mainly on multi-step tasks.

The keyframe channel is the most model-dependent: strongly positive with narration on Claude and Gemini 2.5, but regressing on Gemini 3 due to image-token dilution. Overall, AOI components must be tuned per model.

| Model | Std. | AOI | Δ |
|---|---|---|---|
| Gemini 3 Flash | 36 | 45 | +9 |
| Grok-4.3 | 25 | 65 | +40 |
| Grok-4-fast | 19 | 47 | +28 |

Table 4: Model sensitivity on 100 DynaCU-Bench tasks.

## 7 Model Sensitivity Study

To characterize sensitivity to the underlying model and its inference latency, we evaluate AOI on three additional model configurations (Table 4). Their identifiers are `gemini-3-flash-preview`, `grok-4.3`, and `grok-4-fast-reasoning`; they use the same 1,024-token cap and default sampling.

**Grok models** are primarily limited by high inference latency rather than perception. Grok-4 achieves only +21 pp with AOI due to frequent timeouts. The Grok-4-fast-reasoning and Grok-4.3 configurations achieve larger gains (+28 pp and +40 pp, respectively), indicating that reduced latency allows more of AOI's perceptual benefit to be realized within the task budget.

**Gemini 3 Flash** achieves only a modest +9 pp net gain. A four-way decomposition (Figure 8) reveals a cancellation of effects: audio and the structured scaffold help as usual, but the keyframe stream regresses by −12 pp due to image-token dilution. Dropping keyframes alone recovers +21 pp over the standard baseline, showing the default AOI bundle is mis-tuned for this model.

These cases illustrate that AOI gains are robust but model-specific: the optimal component mix must be chosen per model rather than applied as a fixed bundle.

## 8 Comparison with Streaming Multimodal Baselines

We compare AOI with Gemini Live (`gemini-2.5-flash-native-audio-latest`) (Google 2025) and OpenAI Realtime (`gpt-realtime-2`) (OpenAI 2024). Neither is a CU agent; we stream page audio and screenshots into one session and execute returned tool calls. On AOI's 12-task audio subset, we use each model's highest vision- and function-calling-capable tier, supply no seeds or sampling parameters, and did not retain the output cap.

| Configuration | Pod/Meet/Phone/Intv | Total / 12 |
|---|---|---|
| OpenAI Realtime | 0/0/0/3 | 3 |
| Gemini Live | 0/0/0/0 | 0 |
| AOI full (Claude 4.6) | 3/3/3/3 | **12** |

Table 5: Streaming baselines on 12 audio tasks (3 per family).

Both underperform AOI by a wide margin (Table 5). OpenAI Realtime succeeds only on interview questions answerable from one audio chunk; Gemini Live rarely emits usable tool calls. On five visual controls, both adapters perceive the page and issue valid browser actions (5/5) but complete no tasks; AOI-equipped Claude solves 4/5. Thus, streaming

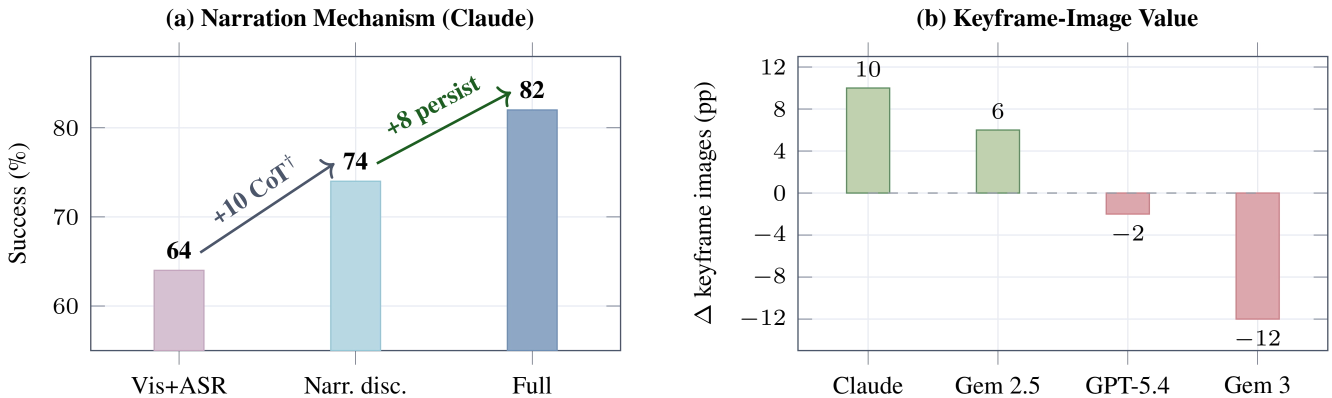


Figure 7: Abalation study that shows the contribution of narration and keyframes. (a) Visual narration improves performance. (b) The value of keyframes is model-dependent.

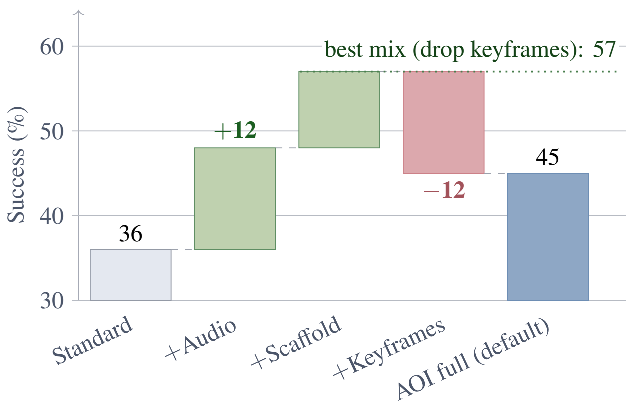


Figure 8: Gemini 3 Flash component decomposition: audio and the scaffold help, while keyframes cost 12 pp.

alone is insufficient, and persistent text for transient observations may be important for later actions.

## 9 Related Work

**Agent interfaces and CU systems.** Action interfaces add commands, code, or grounded elements (Yang et al. 2024; Wang et al. 2024a; Deng et al. 2023); observation work adds trees, HTML, or overlays (Zhou et al. 2023; Yang et al. 2023; Zheng et al. 2024). Yet CU agents (Anthropic 2024a; OpenAI 2025; Google DeepMind 2025; Andreux et al. 2025; Wang et al. 2025a,b; Awadallah et al. 2025; Ye et al. 2025; Song et al. 2025; Simular AI 2025) largely retain one screenshot per step (Sager et al. 2025).

**Temporal and adaptive perception.** Streaming models (Google 2025; OpenAI 2024) couple perception and reasoning; dynamic-GUI benchmarks (Chen et al. 2024; Jang et al. 2024; Chen et al. 2025; Park et al. 2025) expose temporal failures without a reusable observation layer. Video QA selects frames with CLIP (Wang et al. 2024b; Tang et al. 2025; Zohar et al. 2024); adaptive systems (Tan et al. 2024; Jin et al. 2025; Yang et al. 2025; Qian et al. 2025) separate perception. AOI combines online filtering, speech, and persistent memory for existing CU models.

**Structured protocols.** MCP (Anthropic 2024b) and NLWeb (Microsoft 2025) require modified sites; AOI does not.

## 10 Limitations

**Latency and temporal resolution.** CU models take 1–5 s per step; $\sim$3 Hz sampling misses sub-$\sim$300 ms events and real-time games.

**Perception is not reasoning.** Better observations do not fix reasoning, the primary bottleneck on smaller models.

**Audio is speech-only and narration can err.** AOI transcribes speech only, and narrations can propagate recognition errors.

**External validity.** We use a controlled browser with synthesized audio; existing dynamic-GUI benchmarks need reinstrumented harnesses and remain future work.

**Statistical power.** Most configurations have one 100-task run; despite paired McNemar tests, few-point differences warrant caution.

## 11 Conclusion

We treat observation as a separable interface and introduce AOI, an adaptive perception layer compatible with existing CU models. On DynaCU-Bench, AOI raises success by 17–48 percentage points across eight models in the main and open-source replications, making many audio and temporal tasks solvable without retraining. Controls show that the gains arise from both the structured observation record and added perception, rather than prompt formatting alone. Audio unlocks speech-dependent tasks, while visual narration is the largest component and converts transient visual evidence into persistent context. Raw keyframes remain model-dependent and can hurt through image-token dilution, so the channel mix should be tuned for each model. AOI also reduces interaction steps and cloud-model token costs, although observation processing adds latency for faster models. Together, these findings establish observation design as a separate lever for improving CU agents.

**AI disclosure.** We used generative AI for language, code, and formatting; we verified all outputs.

## References


Andreux, M.; et al. 2025. Surfer 2: The Next Generation of Cross-Platform Computer Use Agents. *arXiv preprint arXiv:2510.19949*.

Anthropic. 2024a. Claude Computer Use. https://docs.anthropic.com/en/docs/agents-and-tools/computer-use.

Anthropic. 2024b. Model Context Protocol. https://modelcontextprotocol.io.

Awadallah, A.; et al. 2025. Fara-7B: An Efficient Agentic Model for Computer Use. *arXiv preprint arXiv:2511.19663*.

Chen, D.; et al. 2024. GUI-World: A Video Benchmark and Dataset for Multimodal GUI-oriented Understanding. *arXiv preprint arXiv:2406.10819*.

Chen, L.; et al. 2025. CUA-Suite: 55 Hours of Real Computer-Use Recordings for Agent Benchmarking. *arXiv preprint arXiv:2510.10142*.

Deng, X.; Gu, Y.; Zheng, B.; Chen, S.; Stevens, S.; Wang, B.; Sun, H.; and Su, Y. 2023. Mind2Web: Towards a Generalist Agent for the Web. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Google. 2025. Live API (Gemini Live). https://ai.google.dev/gemini-api/docs/live.

Google DeepMind. 2025. Gemini 2.5 Computer Use. https://deepmind.google/models/gemini/computer-use/.

He, H.; et al. 2024. WebVoyager: Building an End-to-End Web Agent with Large Multimodal Models. *arXiv preprint arXiv:2401.13919*.

Jang, L.; et al. 2024. VideoWebArena: Evaluating Long Context Multimodal Agents with Video Understanding Web Tasks. *arXiv preprint arXiv:2410.19100*.

Jin, Y.; et al. 2025. ScreenLLM: Stateful Screen Schema for Efficient Action Understanding and Prediction. *arXiv preprint arXiv:2503.20978*.

Meituan. 2026. EvoCUA: Evolving Computer Use Agent. https://github.com/meituan/EvoCUA. Accessed 2026-05-14.

Microsoft. 2025. NLWeb: A Conversational Interface for Websites. https://github.com/microsoft/NLWeb.

OpenAI. 2024. Realtime API: Speech-to-speech models with tool use. https://platform.openai.com/docs/guides/realtime.

OpenAI. 2025. Computer-Using Agent (CUA). https://openai.com/index/computer-using-agent/.

OpenAI. 2026. Introducing GPT-5.4. https://openai.com/index/introducing-gpt-5-4/.

Park, J.; et al. 2025. MemGUI-Bench: Benchmarking Memory in GUI Agents. *arXiv preprint arXiv:2509.12233*.

Qian, R.; et al. 2025. Dispider: Enabling Video LLMs with Active Real-Time Interaction via Disentangled Perception, Decision, and Reaction. *arXiv preprint arXiv:2501.03218*.

Qwen Team. 2025. Qwen3-VL Technical Report. *arXiv preprint arXiv:2511.21631*.

Radford, A.; Kim, J. W.; Xu, T.; Brockman, G.; McLeavey, C.; and Sutskever, I. 2023. Robust Speech Recognition via Large-Scale Weak Supervision. In *ICML*. ArXiv:2212.04356.

Sager, P. J.; et al. 2025. A Comprehensive Survey of Agents for Computer Use: Foundations, Challenges, and Future Directions. *arXiv preprint arXiv:2501.16150*.

Simular AI. 2025. Agent S3: An Open-Source Computer-Use Agent. https://github.com/simular-ai/Agent-S.

Song, L.; et al. 2025. CoAct-1: Computer-using Agents with Coding as Actions. *arXiv preprint arXiv:2508.03923*.

Tan, W.; et al. 2024. Cradle: Empowering Foundation Agents Towards General Computer Control. *arXiv preprint arXiv:2403.03186*.

Tang, X.; et al. 2025. Adaptive Keyframe Sampling for Long Video Understanding. *arXiv preprint arXiv:2502.21271*.

Wang, H.; et al. 2025a. UI-TARS-2 Technical Report: Advancing GUI Agent with Multi-Turn Reinforcement Learning. *arXiv preprint arXiv:2509.02544*.

Wang, X.; Chen, Y.; Yuan, L.; Zhang, Y.; Li, Y.; Peng, H.; and Ji, H. 2024a. Executable Code Actions Elicit Better LLM Agents. In *International Conference on Machine Learning (ICML)*.

Wang, X.; et al. 2025b. OpenCUA: Open Foundations for Computer-Use Agents. *arXiv preprint arXiv:2508.09123*.

Wang, Z.; et al. 2024b. VideoTree: Adaptive Tree-based Video Representation for LLM Reasoning on Long Videos. *arXiv preprint arXiv:2405.19209*.

xAI. 2026. Grok 4. https://x.ai/news/grok-4. Accessed 2026-05-14.

Xie, T.; et al. 2024. OSWorld: Benchmarking Multimodal Agents for Open-Ended Tasks in Real Computer Environments. *arXiv preprint arXiv:2404.07972*. NeurIPS 2024.

Yang, H.; et al. 2025. StreamAgent: Towards Anticipatory Agents for Streaming Video Understanding. *arXiv preprint arXiv:2508.01875*.

Yang, J.; Zhang, H.; Li, F.; Zou, X.; Li, C.; and Gao, J. 2023. Set-of-Mark Prompting Unleashes Extraordinary Visual Grounding in GPT-4V. *arXiv preprint arXiv:2310.11441*.

Yang, J.; et al. 2024. SWE-agent: Agent-Computer Interfaces Enable Automated Software Engineering. *arXiv preprint arXiv:2405.15793*.

Ye, J.; et al. 2025. Mobile-Agent-v3: Fundamental Agents for GUI Automation. *arXiv preprint arXiv:2508.15144*.

Zhang, A. L.; et al. 2025. VideoGameBench: Can Vision-Language Models complete popular video games? *arXiv preprint arXiv:2505.18134*.

Zhang, C.; et al. 2023. A Simple LLM Framework for Long-Range Video Question-Answering. *arXiv preprint arXiv:2312.17235*.

Zheng, B.; Gou, B.; Kil, J.; Sun, H.; and Su, Y. 2024. GPT-4V(ision) is a Generalist Web Agent, if Grounded. In *International Conference on Machine Learning (ICML)*.

Zhou, S.; et al. 2023. WebArena: A Realistic Web Environment for Building Autonomous Agents. *arXiv preprint arXiv:2307.13854*. ICLR 2024.

Zohar, O.; et al. 2024. Apollo: An Exploration of Video Understanding in Large Multimodal Models. *arXiv preprint arXiv:2412.10360*.